

VIRAJ RAOSAHEB PATIL

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Professional Summary

AI engineer (B.Tech final year) skilled in LLMs, RAG, NLP, and full-stack AI integration. Comfortable across the stack — from embedding pipelines and vector search to REST APIs and React frontends. Seeking AI/ML Engineering roles to build real-world GenAI solutions.

Technical Skills

Languages: Python, SQL

Generative AI & LLMs: LangChain, LlamaIndex, OpenAI API, Hugging Face, RAG Pipelines, Prompt Engineering, Tool Calling, Embeddings, FAISS, ChromaDB

ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, RNNs, DNNs, Model Evaluation, Fine-tuning

NLP: spaCy, BioBERT, BERT/GPT Transformers, NER, Text Classification

Backend & Infrastructure: FastAPI, Flask, React.js, REST APIs, Firebase Firestore, Docker (basic), GitHub Actions (CI/CD)

Data & Databases: NumPy, Pandas, Matplotlib, Seaborn, MySQL, ChromaDB, Firebase

Tools: Git, GitHub, Jupyter Notebook

Projects

DocTalk - RAG-Powered PDF Chat Application | *Python, LangChain, FAISS, FastAPI, React.js* 🔄 **Ongoing**

- Planning a production-grade RAG PDF Q&A system for processing **100+ page documents** - LangChain · FAISS · FastAPI backend with React.js frontend, integrating OpenAI/Hugging Face LLMs for context-aware, cited document understanding.
- Designing an end-to-end retrieval pipeline with chunking, embedding, and vector similarity search to support multi-turn natural language Q&A with source-referenced answers.

SmartAgriAI - Soil Analysis & Crop Recommendation Engine | *Python, TensorFlow, Keras, Scikit-learn* 🔄 **May 2025**

- Built an end-to-end ML pipeline on **5,000+ soil samples** using KNN imputation and a **multi-task DNN** (TensorFlow/Keras), achieving **99.39% accuracy** - outperforming SVM (94.2%) and Random Forest (96.8%) baselines.
- Deployed as a **Flask web application** with full HTML/CSS frontend; generates crop-specific fertilizer schedules (NPK + micronutrients) and season-aware timelines for **10+ crop types** across Kharif, Rabi, and Zaid seasons.

NavigateU - AI-Powered Accessible Navigation System | *React.js, Firebase, Leaflet.js, NLP* 🔄 **Sep 2025**

- Built an accessibility navigation platform with NLP-based voice query parsing and real-time discovery of **50+ wheelchair-accessible facilities** across mapped urban areas; AI recommendation engine ranks results by proximity and accessibility score.
- Integrated Firebase Firestore for live data sync and Leaflet.js for interactive map rendering; reduced average facility search time by providing instant AI-ranked results over manual map browsing.

Explainable AI for Clinical Text-to-ICD-10 Mapping | *Python, BioBERT, TensorFlow, spaCy* 🔄 **June 2026**

- Building an NLP pipeline for automated ICD-10/CPT medical code prediction from clinical notes using **BioBERT fine-tuning**, RNNs, and NER-based medical entity extraction - benchmarking F1-score on i2b2 and clinical note datasets.
- Applying attention visualization and explainability techniques (SHAP-style token importance) to make model predictions interpretable and audit-ready for clinical use cases.

Certifications

- Gen AI Launchpad: From Beginner to Builder** — Innomatics Research Labs
- Deloitte Australia** — Data Analytics Job Simulation, Forage
- SQL (Basic & Intermediate)** — HackerRank Skills Certification
- GitHub Actions CI/CD Fundamentals** — Microsoft Learn

Education

Nutan College of Engineering and Research <i>Bachelor of Technology in Computer Science (Artificial Intelligence)</i>	Pune, India <i>2026</i>
Shri Shivaji Vidya Mandir, Aundh <i>12th HSC – Maharashtra State Board</i>	Pune, India <i>2022</i>

Extracurricular Activities

- **Referee · First Tech Challenge Robotics Competition**, Baner, Pune — coordinated rules enforcement across **20+ student teams**, demonstrating cross-functional communication and composure under pressure.
- **NSS Team Lead** — directed a **30-volunteer team** across 2 community initiatives including Blood Donation Camp and Wari Cleanliness Drive; collected **200+ plastic bottle waste**, demonstrating ownership and leadership.

Languages

English | Marathi | Hindi | Kannada